

Inverse Design of Light-Trapping Layers

Andreas Nymand

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Abstract

This report details the surrogate optimization and active learning approaches used to inversely design light-trapping layers.

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1 Introduction

Introduce the problem of light-trapping layers here [?].

2 Methods and theory

2.1 Rigorous coupled wave analysis

This subsection will introduce the concept of rigorous coupled wave analysis (RCWA), a method in numerical photonics suited well for stacked periodic structures in the x and y axis with a period Λ_x, Λ_y respectively, and where each layer is homogeneous in the z-direction [?]. The structure's optical properties are completely described by the relative permittivity and relative permeability of each layer, meaning that each structure will have a geometry described by

$$\epsilon(x, y, z)_N = \epsilon(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z} \quad (1)$$

and

$$\mu(x, y, z)_N = \mu(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z}, \quad (2)$$

where m and n are integers and the subscript N denotes the N 'th layer of the structure. For the entirety of this project, the designed fourier patterned light trapping layers will be designed using silicon (Si), Titanium Dioxide (TiO_2) and Silicon Nitride (Si_3N_4), which are all nonmagnetic, so the relative permeability $\mu = 1$. The equations solved for are of course the Maxwell equations, namely Faraday's law

$$\nabla \times \mathbf{E} = -\mu_0 \mu(x, y) \mathbf{H} \quad (3)$$

and Ampere's law

$$\nabla \times \mathbf{H} = \epsilon_0 \epsilon(x, y) \mathbf{E} \quad (4)$$

2.2 Machine Learning Surrogates

2.3 Optimization Strategies

3 Results

Present the performance of the surrogate models and the final inverse design outcomes.

Figure 1: Placeholder for a results figure.

4 Discussion

Discuss the results and what worked/didn't work.

5 Outlook

Discuss future work.

6 Conclusion

Summarize the findings.